

## Other Capital Structure Issues

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This appendix discusses alternative models of capital structure and credit rating estimations. These models offer some interesting insights but tend to be less useful in practice for designing a company's capital structure. Finally, the appendix shows the similarities and differences between widely used credit ratios such as leverage, coverage, and solvency.

### PECKING-ORDER THEORY

An alternative to the view that there are trade-offs between equity and debt is a school of thought in finance theory that sees a pecking order in financing.<sup>1</sup> According to this theory, companies meet their investment needs first by using internal funds (from retained earnings), then by issuing debt, and finally by issuing equity. One of the causes of this pecking order is that investors interpret financing decisions by managers as signals of a company's financial prospects. For example, investors will interpret an equity issue as a signal that management believes shares are overvalued. Anticipating this interpretation, rational managers will turn to equity funding only as a last resort, because it could cause the share price to fall. An analogous argument holds for debt issues, although the overvaluation signal is much smaller because the value of debt is much less sensitive to a company's financial success.<sup>2</sup>

<sup>1</sup> See G. Donaldson, "Corporate Debt Capacity: A Study of Corporate Debt Policy and the Determination of Corporate Debt Capacity" (Harvard Graduate School of Business, 1961); and S. Myers, "The Capital Structure Puzzle," *Journal of Finance* 39, no. 3 (1974): 575–592.

<sup>2</sup> An exception is, of course, the value of debt in a financially distressed company.

According to the theory, companies will have lower leverage when they are more mature and profitable, simply because they can fund internally and do not need any debt or equity funding. However, evidence for the theory is not conclusive. For example, mature companies generating strong cash flows are among the most highly leveraged, whereas the pecking-order theory would predict them to have the lowest leverage. High-tech start-up companies are among the least leveraged, rather than debt loaded, as the theory would predict.<sup>3</sup> Empirical research shows how the signaling hypotheses underlying the pecking-order theory are more relevant to financial managers in selecting and timing specific funding alternatives than for setting long-term capital structure targets.<sup>4</sup> Surveys among financial executives confirm these findings.<sup>5</sup>

## MARKET-BASED RATING APPROACH

Alternative metrics to credit ratings have been developed based on the notion that equity can be modeled as a call option on the company's enterprise value, with the debt obligations as the exercise price.<sup>6</sup> Using option valuation models and market data on price and volatility of the shares, these approaches estimate the future probability of default—that is, the probability that enterprise value will be below the value of debt obligations.<sup>7</sup> The advantage is that all information captured by the equity markets is directly translated into the default estimates. Traditional credit ratings tend to lag changes in a company's performance and outlook because they aim to measure credit quality "through the cycle"<sup>8</sup> and are less sensitive to short-term fluctuations in quality.

The disadvantage of market-based ratings is that no fundamental analysis is performed on the company's underlying business and financial health. If

<sup>3</sup> See M. Barclay and C. Smith, "The Capital Structure Puzzle: The Evidence Revisited," *Journal of Applied Corporate Finance* 17, no. 1 (2005): 8–17; and M. Baker and J. Wurgler, "Market Timing and Capital Structure," *Journal of Finance* 52, no. 1 (2002): 1–32.

<sup>4</sup> See also A. Hovakimian, T. Opler, and S. Titman, "The Debt-Equity Choice," *Journal of Financial and Quantitative Analysis* 36, no. 1 (2001): 1–24, for evidence that the pecking-order theory predicts short-term movements in corporate debt levels but that long-term changes are more in line with the trade-offs discussed earlier in this section.

<sup>5</sup> J. Graham and H. Campbell, "How Do CFOs Make Capital Budgeting and Capital Structure Decisions?" *Journal of Applied Corporate Finance* 15, no. 1 (2002): 8–23.

<sup>6</sup> This is because equity is a residual claim on the enterprise value after payment of principal and interest for debt. It has value only to the extent that enterprise value exceeds debt commitments. See R. Merton, "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *Journal of Finance* 29 (1974): 449–470; or for an introduction, R. Brealey, S. Myers and F. Allen, *Principles of Corporate Finance*, 13th ed. (New York: McGraw-Hill, 2020), chap. 23.

<sup>7</sup> See P. Crosbie and J. Bohn, "Modeling Default Risk" (Moody's KMV White Paper, December 2003).

<sup>8</sup> See E. Altman and H. Rijken, "How Rating Agencies Achieve Rating Stability," *Journal of Banking and Finance* 28, no. 11 (2004): 2679–2714.

equity markets have missed some critical information, the resulting estimates of default probability do not reflect their omission. As discussed in Chapter 7, markets reflect company fundamentals most of the time, but not always. When they do not, the market-based rating approaches would incorrectly estimate default risk as well.<sup>9</sup>

## LEVERAGE, COVERAGE, AND SOLVENCY

The leverage measure used in the academic literature is typically defined as the market value of debt ( $D$ ) over the market value of debt plus equity ( $E$ ):

$$\text{Leverage} = \frac{D}{D+E}$$

This ratio measures how much of the company's enterprise value is claimed by debt holders and is an important concept for estimating the benefits of tax shields arising from debt financing. It is therefore also a crucial input in calculating the weighted average cost of capital (WACC; see Chapter 15 on capital structure weights).

Compared with coverage ratios such as earnings before interest, taxes, and amortization (EBITA) to interest, leverage ratios suffer from several drawbacks as a way to measure and target a company's capital structure. First, companies could have very low leverage in terms of market value but still be at a high risk of financial distress if their short-term cash flow is low relative to interest payments. High-growth companies usually have very low levels of leverage, but this does not mean their debt is low-risk. A second drawback is that market value can change radically (especially for high-growth, high-multiple companies), making leverage a fast-moving indicator. For example, during the stock market boom of the late 1990s, several European telecom companies had what appeared to be reasonable levels of debt financing in terms of leverage. Credit providers appeared willing to provide credit even though the underlying near-term cash flows were not very high relative to debt service obligations. But when the companies' market values plummeted in 2001, leverage for these companies shot up, and financial distress loomed. Thus, it is risky to base a capital structure target on a market-value-based measure.

This does not mean that leverage and coverage are fundamentally divergent measures. Far from it: they actually measure the same thing but over different time horizons. For ease of explanation, consider a company that has

<sup>9</sup> See Crosbie and Bohn, "Modeling Default Risk," 23.

no growth in revenues, profit, or cash flows. For this company, it is possible to express the leverage and coverage as follows:<sup>10</sup>

$$\begin{aligned}\text{Leverage} &= \frac{D}{D+E} \\ &= \frac{\text{Interest}_1 + \text{PV}(\text{Interest}_2) + \dots + \text{PV}(\text{Interest}_\infty)}{\text{NOPAT}_1 + \text{PV}(\text{NOPAT}_2) + \dots + \text{PV}(\text{NOPAT}_\infty)} \\ \text{Coverage} &= \frac{\text{EBITA}}{\text{Interest}} = \frac{1}{(1-T)} \times \frac{\text{NOPAT}}{\text{Interest}}\end{aligned}$$

where  $D$  = market value of debt  
 $E$  = market value of equity  
 $\text{NOPAT}_t$  = net operating profit after taxes in year  $t$   
 $\text{Interest}_t$  = interest expenses in year  $t$   
 $T$  = tax rate

The market value of debt captures the present value of all future interest payments, assuming perpetual rollover of debt financing. The enterprise value ( $E + D$ ) is equal to the present value of future NOPAT, because depreciation equals capital expenditures for a zero-growth company. A leverage ratio therefore measures the company's ability to cover its interest payments over a very long term. The problem is that short-term interest obligations are what mainly get a company into financial distress. Coverage, in contrast, focuses on the short-term part of the leverage definition, keeping in mind that NOPAT roughly equals  $\text{EBITA} \times (1 - T)$ . It indicates how easily a company can service its debt in the near term.

Both measures are meaningful, and they are complementary. For example, if market leverage were very high in combination with strong current interest coverage, this could indicate the possibility of future difficulties in sustaining current debt levels in, for example, a single-product company faced with rapidly eroding margins and cash flows because the product is approaching the end of its life cycle. Despite very high interest coverage today, such a company might not be given a high credit rating, and its capacity to borrow could be limited.

Solvency measures of debt over book value of total assets or equity are seldom as meaningful as coverage or leverage. The key reason is that these book value ratios fail to capture the company's ability to comply with debt service requirements in either the short term or the long term. Market-to-book ratios

<sup>10</sup> The simplifying no-growth assumption is for illustration purposes only. For a growing company, the same point holds.

can vary significantly across sectors and over time, making solvency a poor proxy for long-term ability to service debt.

Solvency becomes more relevant in times of financial distress, when a company's creditors use it as a rough measure of the available collateral. Higher levels of solvency usually indicate that debt holders stand better chances of recovering their principal and interest due—assuming that asset book values are reasonable approximations of asset liquidation values. However, in a going concern, solvency is much less relevant for deciding capital structure than coverage and leverage measures.

